Q U A R K - H A W K I N G B R I D G E

The Quark-Hawking Bridge. Volume II: The Appendix

constants, models, campaigns C1–C11, data, runbook, FAQ

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Part I. The constants registry

The single source of truth is qhb/constants.py, three parts. Part A: CODATA-2018 and the cosmological/QCD anchors (T_CMB = 2.72548 K, T_QCD = 155.16 MeV in exact conversion, eta = 6.1e-10, the Wu anchor 0.9 marked "approximate"). Part B: the CHB inheritance — the spinor phases and the six correction quanta (exact), the born-ensemble members f~ = 0.2877 / 0.2684 / 1.0 (inherited from CHB campaign C1), the parent package's T10 drift-law data. Part C: the committed verdicts of campaigns C1–C11, regenerated by the run_c*.py scripts. Since v1.2.0 part C also carries the phase-map block (C9), the ablation (C10) and the vortex anchor (C11).

Part II. The G-model of the gap

Geometry: a Dirichlet chain of N sites with the parabolic well $q_n = (\kappa/2)x_n^2 - q_{pk}$ and the continuum-consistent hopping $J = 1/h^2$. Reduction: the two lowest levels → a two-band system with the splitting Δ_0. Drive: $H(t) = (\Delta_0/2)\sigma_z + v(t)\sigma_x + w(t)\sigma_z$, v — the Berry modulation (transverse), w — the braking one (longitudinal), the tone frequencies $(2,3,7)\cdot\Omega_0$, the septinial phases. Floquet: U(T0) built from M = 1920 piecewise-constant exact 2x2 exponentials; the quasi-energies come from the eigenphases with folding; the gap is the smaller of the two intervals.

The validity boundary of the two-band reduction is $\Delta_0 \le \Omega_0/2 = 1$; at $q_{pk} < 0.2$ the static splitting crosses the band edge and the model honestly leaves its regime — campaign C9 excludes this branch from the map rather than displaying it.

Part III. The trio engine

engine/solver.py — a vectorised RK4 over an ensemble: three equal masses, softened Newtonian gravity ($\varepsilon^2 = 1e\text{-}2$), step dt = 2e-3, horizon $t_{end} = 12$, escape radius $R_{esc} = 12$, early freezing of crossers. The wall: $U(r,t) = V_0(1 + \cos \Omega t)\cdot\max(0, (r - R_b)/w)^2$, $R_b = 8$, w = 1, p = 2. Ensemble: 300 trials in the band E in $[-0.6, -0.05)$, seed 20261010. The validation anchor is the Kepler ellipse e = 0.5: the period relative error 1.3e-9.

Since v1.2.0 the engine carries the exact static barrier (T13): at $\Omega = 0$ the energy is conserved and every solution obeys the bound $r_i \le R_b + w\cdot\sqrt{(E_0 + 3/\sqrt{\varepsilon^2})/(2V_0)}$; escape is impossible at $V_0 > V^* = 0.9359375$ on the committed band. The oscillating wall does work (median |dE| = 0.146 on the (0.8, 2) cell) — the seal is finite-time, and both marks travel with the numbers.

Part IV. The W-model of the Wu door

Linear two-rate kinetics: births Gamma_b, annihilations Gamma(1 −/+ alpha), freeze-out at t_f = 1/H. The closed form of the residual r(alpha, x) is theorem T7 of the theorems volume. The stochastic realisation: N_± ~ Poisson with thinned survival, 200,000 realisations per cell, seed 20261010. The C8 pairwise kinetics replaces the linear mean field; the T7 closed form does not extend — an exact structural negative.

Part V. Campaigns C1–C8 (v1.0–v1.1)

The eight campaigns of the first cycle: C1 — statics and the gap law (T2/T3); C2 — the trap: Gascheau, the 9x4 grid, Lyapunov (T4–T6); C3 — the Wu door (T7); C4 — the spectral floor (T9); C5 — the drift law (T10); C6 — the three-branch physical mapping; C7 — the anharmonic trio with the confining core; C8 — the pairwise door. All verdicts are deterministic, carry project_version 1.1.0 and are re-verified by the C7 regression gate and the audit.

Part VI. Campaigns C9–C11 and the Evidence Engine (v1.2.0)

C9 — the phase map. The 9x41 grid (q_pk in {0.2…1.0} under the regime bound Delta_0 ≤ Omega_0/2; beta in [0, 10] step 0.25, M = 1920). Classification against the detector floor 2/N = 0.03125: all 369 cells are OPEN — no closure on the searched range (a finite-range statement, beta_max = 10 is part of the claim). The narrowing valley: beta* = 3.3583, Delta_F,min = 0.181777 at the committed geometry; refinement reproducibility dev 0.0002 < 0.05; N-stability < 0.001. The counterexample: the quadratic closure forecast beta_c = 2.9910 is falsified — the measured 0.1978 sits 6.3x above the floor.

C10 — the competing models. The factorial ablation: 8 sector-mask configurations; the causal shares of |B2| — A 91.95%, B 8.00%, C 0.049%; additivity — max cross 1.3e-4; the braking-only configuration gives B2 = 0.0 exactly; the b-flip evenness holds with machine zeros on all seven transverse configurations. The out-of-window forecast (fit on [0, 0.1], test on {0.3…0.8}): the error share 0.3% → 2.3%, the residual power 4.07 — the quartic wing quantified as a forecast failure, not as a refutation of T3.

C11 — the vortex anchor and the static barrier. The own classical layer qhb/vortex.py: the Kirchhoff equations, RK4 dt = 1e-3, four invariants. T11: rigid rotation to 1.5e-14 over three full turns, the linearization spectrum {0, 0, ±i omega, ±i omega} (the exact analytic Jacobian, max|Re| = 2.8e-17), the symbolic polynomial $\lambda^2(\lambda^2 + \omega^2)^2$ — dev 0 (SymPy, IVP11). T12: the Aref manifold (1, 1, −1) — machine-zero invariants at 50 digits, drift ≤ 1.5e-12, r_max = 2.7654 << 12 — bounded without any wall. T13: the static barrier, V* = 0.9359375, 0/300 escapes, margins ≥ 2.76.

The Evidence Engine. results/claims_registry.json — 27 claims in five statuses (exact / measured-law / honest-negative / hypothesis / comparison) with five-field uncertainty budgets and falsification criteria; the qhb.claims validator, a test pin. The prose registry — results/CLAIMS.md.

Part VII. The theorems register (T1–T13 summary)

Table 7.1 — the thirteen theorems of the package

No.	theorem	status	campaign
T1	the six correction quanta (exact maps)	exact	C1/validate
T2	selection rules: evenness, invisibility	exact + data	C1, C10
T3	the gap law (window [0, 0.2])	measured law	C1
T4	the Gascheau criterion: instability	exact (negative)	C2, IVP06
T5	the free trio unbinds	honest negative	C2
T6	the wall seals; bound chaos	measured law	C2, C7
T7	the Wu door: the closed form	exact in the toy	C3
T8	the QCD contact: M_QCD = 6.814e10 kg	comparison	C6
T9	the spectral floor (2/N)	detector law	C4
T10	the gap-drift law (h^2, channel −0.209)	measured law	C5
T11	the stability contrast (vortices vs Newton)	exact	C11, IVP11
T12	the invariant wall (the Aref manifold)	measured	C11
T13	the static barrier (V* = 0.9359375)	exact for the engine	C11

Part VIII. The bridge and the two contact points

The bridge master table — 15 rows (3 members of the f~ ensemble × 5 input scales: Planck, QCD, asteroid 1e12 kg, CMB, solar) × 23 Schwarzschild-thermodynamics columns. The two contact points: QCD — T_H = 155.16 MeV selects M = 6.814006e10 kg (compared with the literature primordial holes of the QCD epoch ~1e11 kg, ratio 0.68, comparison only); CMB — the inherited contact 4.5016e22 kg. The gap mapping: f_gap = Delta_0·k_B·T/h; at

the QCD contact the gap quantum is 81.1 MeV — the same decade as Lambda_QCD (ratio 4.09), a comparison, not a derivation.

Part IX. The data dictionary

data/qhb/ — 32 derived tables rebuilt byte-for-byte by scripts/rebuild_data.py from the verdicts (a test pin). The main ones: bridge_master.csv, qhb_constants.csv, the scale ladders, gap_scan_window.json, gap_per_sector.json, trap_grid.csv, trap_lyapunov.json, wu_residual_grid.csv, floor_ladder.csv, drift_ladder.csv, mapping_branches.csv, anharmonic_*, pairwise_*, and the new v1.2.0 tables: phase_map.csv (369 cells), phase_valley.csv, phase_counterexample.json, ablation.csv, forecast.csv, ablation_additivity.json, barrier_cells.csv, vortex_anchor.json.

Part X. The figure catalogue

The 300-dpi figures: 10 architectures × RU/EN × light/dark = 40 files (figures/). The new v1.2.0 ones: fig_phase (the valley, the floor, the counterexample), fig_ablation (the causal map), fig_vortex (the Aref manifold + the static barrier). The 600-dpi hires: 25 architectures × RU/EN × themes = 100 files (figures_hires/), including hq_21–hq_25 (the valley, the causal map, the barrier, the phase-plane heatmap, the ablation zoom). All built through scripts/figstyle.py — ink and gold, constrained_layout.

Part XI. The multilang matrix

The twins of the closed-form core: JavaScript, C, C++ — 14/14 PASS each (v1.2.0 adds the vortex Lagrange rate 3/(2 pi) to the 13-check block); the reference outputs are committed (multilang/*/reference_output.txt) and pinned by tests. Go, Rust, Julia — source twins with the same block. Nothing is downloaded, nothing is fitted.

Part XII. The reproduction runbook

```
pip install -e ".[dev]"        # install
make test                     # 94 pins
make validate                 # 9/9 mpmath (50 digits)
make claims                   # 27 claims, the schema
make data                     # 32 tables from the verdicts
make figures                  # 40 figures @300dpi
python3 scripts/hires_figures.py  # 100 hires @600dpi
make c1 c3 c4 c5 c6 c9 c10 c11  # the fast campaigns (~65 s total)
make c2                       # ~3.5 min (the trap)
make c7                       # ~5 min (the anharmonic trio)
bash independent_verification/RUN_ALL.sh   # the audit IVP-11: 11/11
```

All eleven campaigns are deterministic (the single seed 20261010; C6, C8, C9, C10 are fully deterministic closed forms/discretizations): a re-run returns the physics bit-for-bit, only the wall times shift. The heavy campaigns C2/C7 re-run in minutes; the rest take seconds.

Part XIII. The roadmap

Closed: C6–C8 (v1.1.0), C9–C11 + the Evidence Engine + IVP-11 (v1.2.0). Open: the engine modernisation (adaptive stepping at close encounters — an AMR-style engine would sharpen the escape statistics); the adaptive experiment selection (the C9 valley locus is the natural seed for spending the next CPU-hours where the phase-boundary uncertainty is largest); the interactive delivery (an in-browser solver front-end — the core is already pure functions).

Part XIV. FAQ

Does the package prove the Yang-Mills hypothesis? No. Every theorem is a machine fact about a committed toy model; the mapping to physical scales is a marked hypothesis and comparisons. The package honestly approaches the problem from the toy side: the gap never closes on the whole searched map (C9), the drive mechanism is causally decomposed (C10), the trap received its exact stratification (T13) and the classical contrast (T11).

Why the vortex layer? The classical three-vortex problem (Kirchhoff 1876) is the oldest reduction of fluid dynamics to a few-body system, with four analytic invariants and an EXACT contrast to the Newtonian Gascheau instability: the same triangle, a different interaction structure — a different stability. This is the strongest analytic argument of the package in favour of the H4 link: the trap is a property of the interaction, not of "three-ness". The implementation follows the classical equations with the package's own code paths, with the symbolic proof of the characteristic polynomial.

What does "machine-readable or it did not happen" mean? Every numerical claim traces to a committed JSON/CSV verdict with an address; console output is not an artifact. The claims registry binds every load-bearing statement to its status, uncertainty budget and falsification criterion; quoting obliges carrying the status mark.

How to reproduce? The runbook of part XII: install, test, validate, campaigns, audit — everything in one command on a clean environment, without hand-prepared intermediate tables.